Dylan Thornburg

SANTA CLARA UNIVERSITY	ELEN 153	Shoba Krishnan
Laboratory #5: Hierarchical Design: Two-Input XOR Gate		

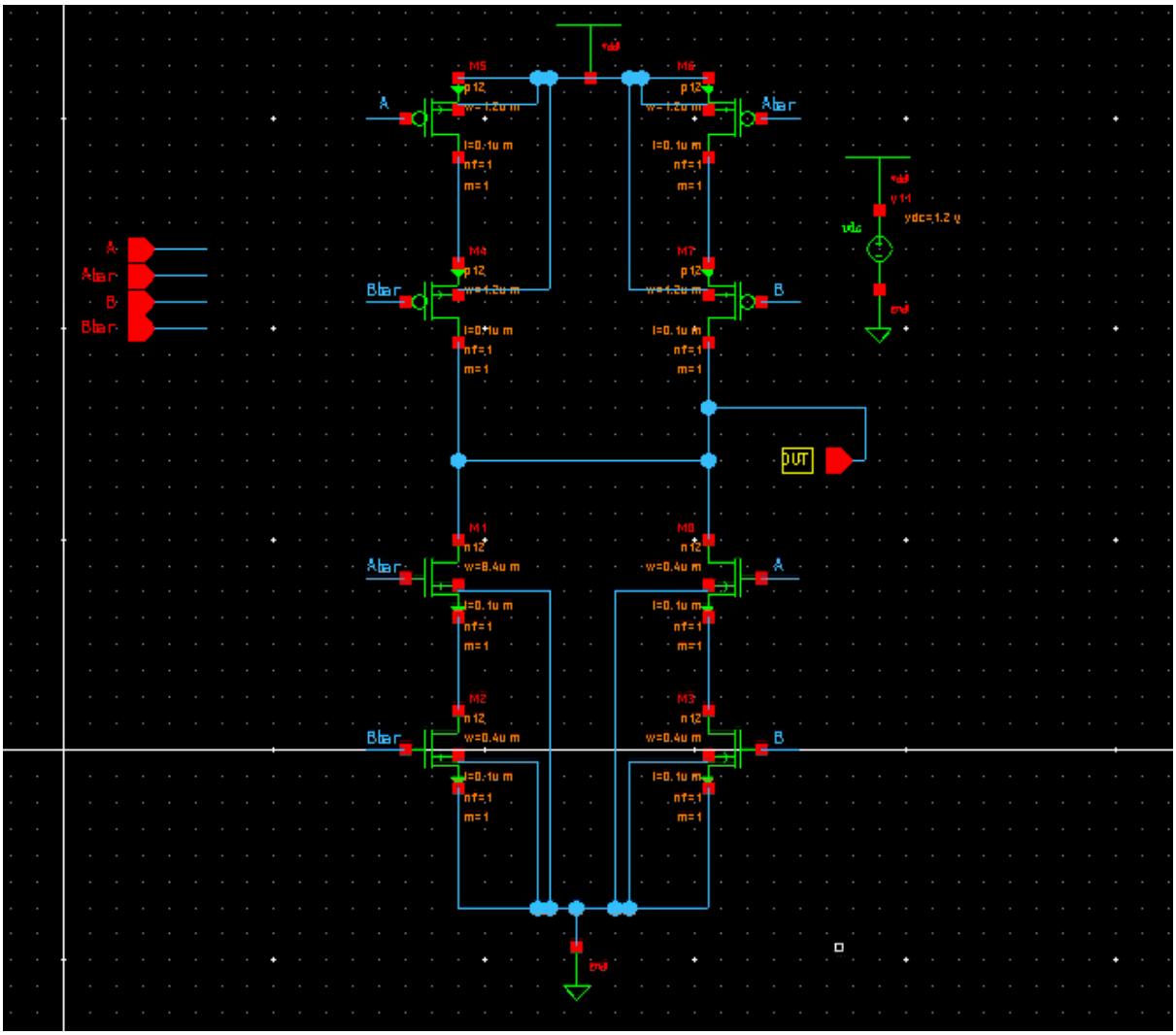
I. OBJECTIVES

- To create a schematic and symbol for a two input XOR gate
- To perform simulations to verify the XOR's functionality.

II. LAB PROCEDURE

PART A: XOR Logic schematic and Symbol

- Draw the transistor level schematic of an XOR Logic gate using Synopsys Custom Compiler. Name your input pins: A, B, Abar, Bbar. Name your output pin as OUT.

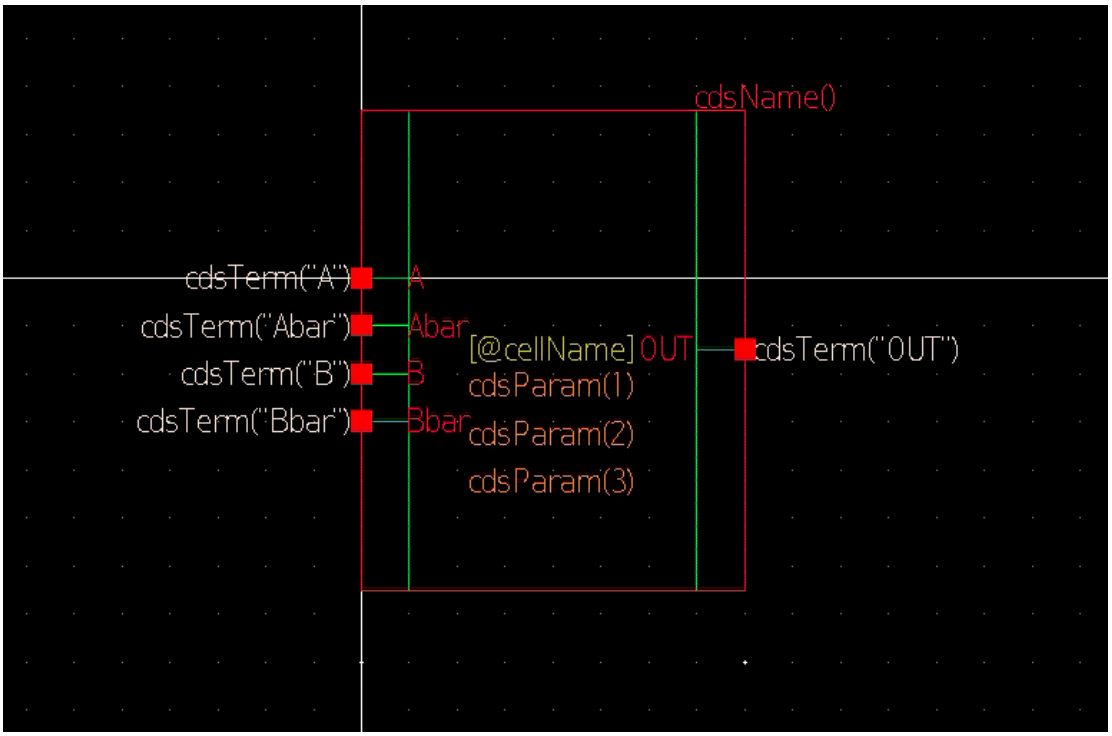


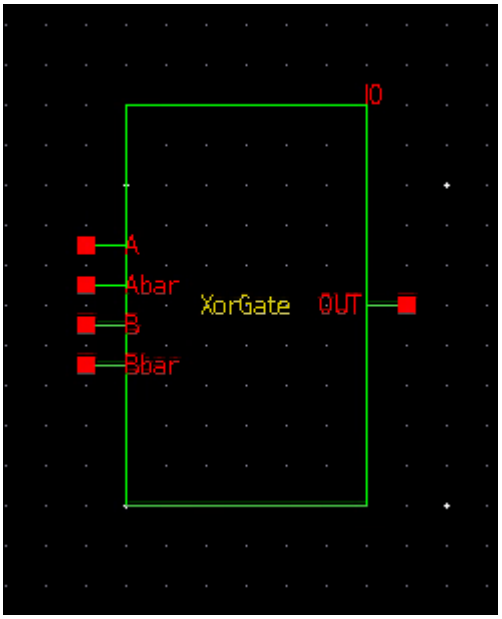
● For the PMOS and NMOS device aspect ratios (W/L), size the devices to match the rise and fall time of the inverter created in previous labs.

Question: Explain the transistor sizing of above XOR in your lab report?

The Inverter had a (W/L)p of 1.2/.1 and a (W/L)n of .4/.1 so I just copied these to ensure similar delay times.

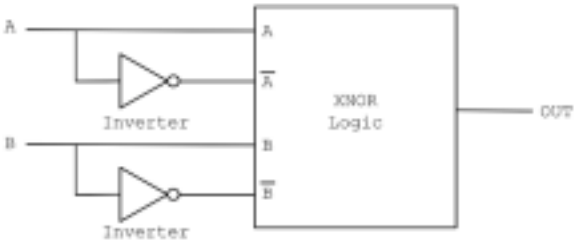
● A symbol will be automatically generated for your schematic. Do not make any changes to your XOR symbol. It should look like default rectangle as seen below.

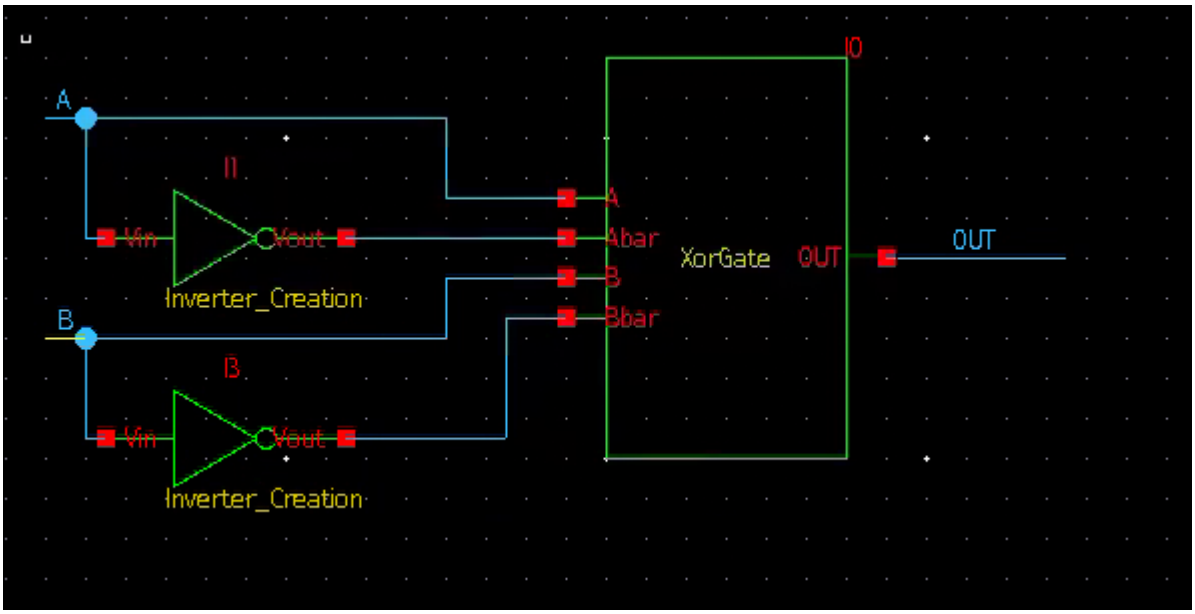




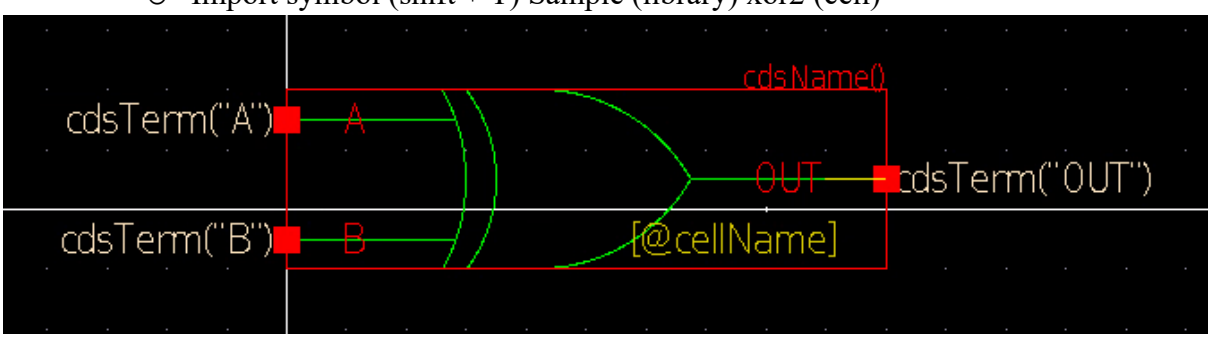
<u>PART B</u>: Four Inputs to Two Inputs using Inverters (Hierarchy)

● Using the above XOR Logic symbol, and the inverters symbol from the previous labs, create a new schematic of a two input XOR Gate as shown below picture. This is a hierarchical schematic that uses previous logic gates you have made.





- Now, create a symbol by replacing the default rectangle with XOR as below.
    - Import symbol (shift + T) Sample (library) xor2 (cell)

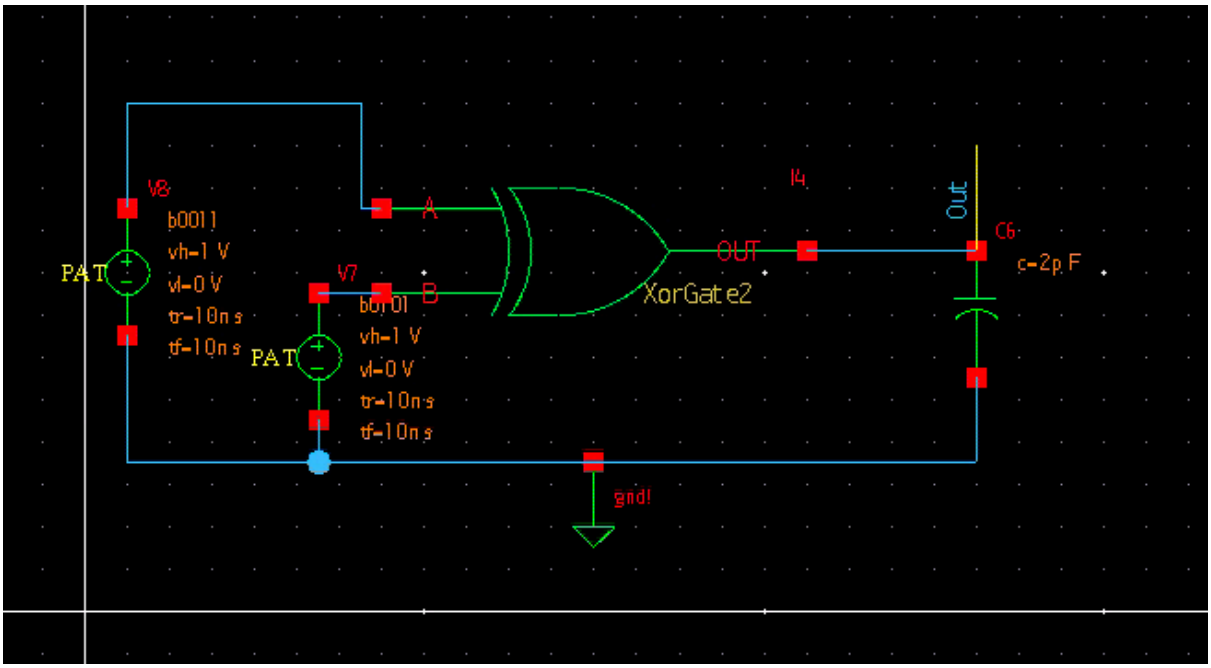


<u>PART C</u>: Transient Simulation

In this section, we will perform a transient simulation to verify the XOR's functionality.

- Create a new test-bench to simulate the XOR gate:
- For inputs (A,B) of the XOR, connect individual Pattern sources **Vpat** from **analoglib**.

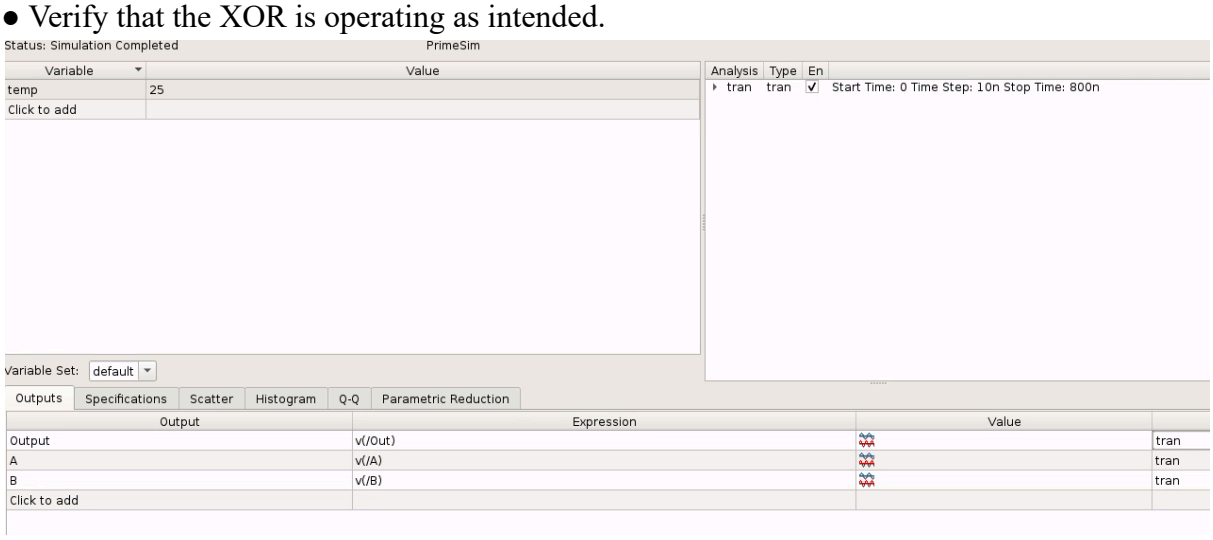
● Setup the patterns in such a way it (00, 01, 10, 11) are produced. (ex. Vpat for A; **data** = b0011). Set the **sample time** as 200n and the **rise/fall time** as 10n ● Connect a load capacitance at OUT: CLOAD = 2pF

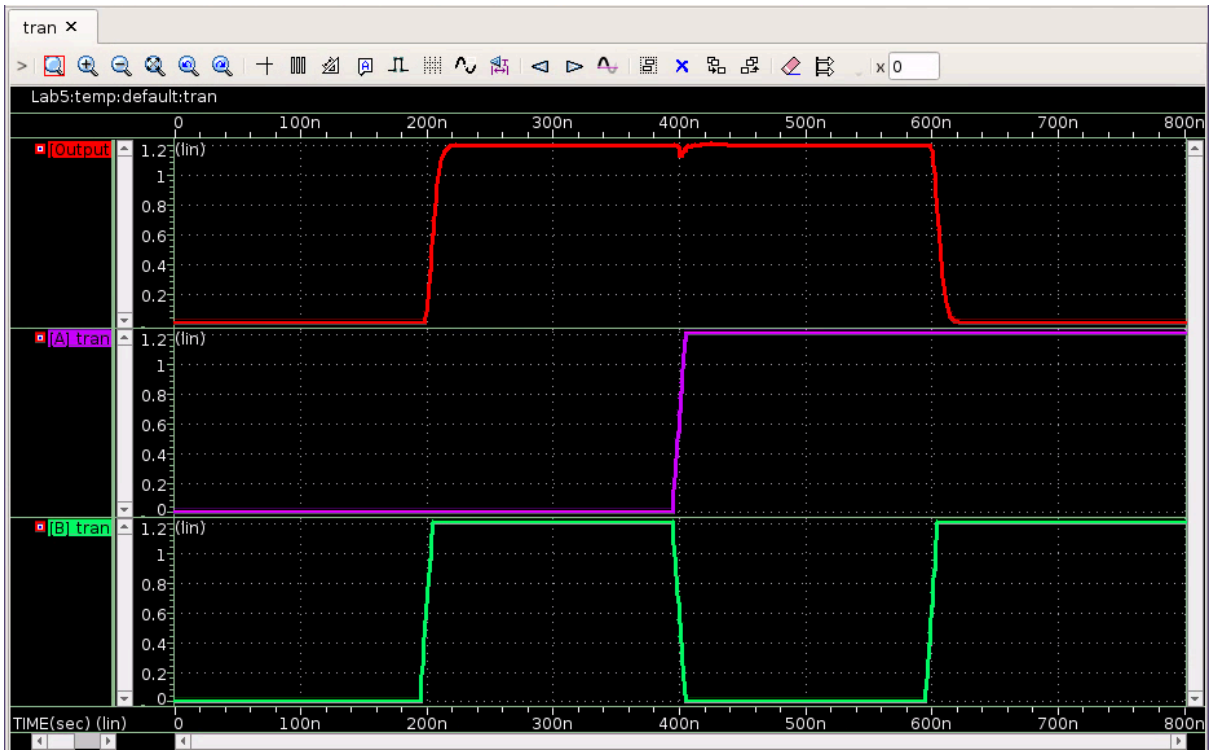


I CHANGED VOLTAGE INPUTS TO 1.2V FOR SIM BTW

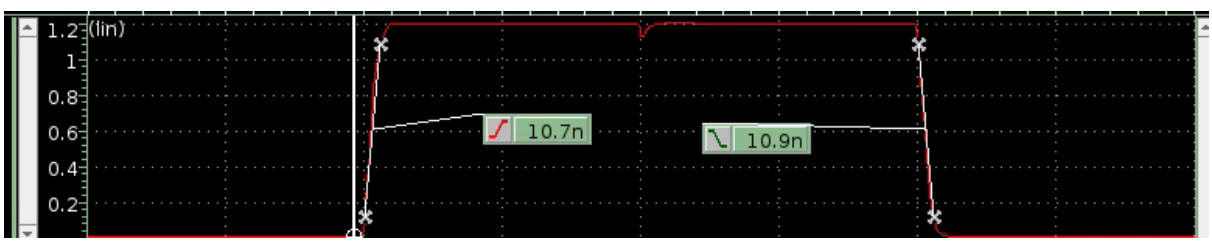
● Use SAE to setup a transient simulation with **time step** 10n, **stop time** 800n and plot all the inputs and outputs.

● Verify that the XOR is operating as intended.





Question: Measure the Rise and Fall time of the output and include the picture with the visible measurements.



Very close and practically the same (but not quite).

III. <u>REPORT</u>

Write a short laboratory report that details all the work done. Describe the objective and procedures of this lab with your own words. The lab report should contain the following:

a) All schematics and testbenches used in your lab.
b) Simulation setup and all waveforms.
c) Answer any questions in the lab assignment
d) Conclusions